

What theories of consciousness commit to, and what they leave unmeasured: an audit of stated predictions after the first adversarial test

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Running head. What theories of consciousness commit to

Abstract

The first preregistered adversarial test of consciousness theories compared two theories on three predictions about the cortical signature of conscious content: on the temporal profile they excluded each other's value and were separated; on the other two each failed its own criterion. We ask what the theories have committed to beyond that signature, and what they have left unmeasured. A purposive audit coded the stated predictions of 12 accounts against 10 measurable quantities (120 cells), registered 27 stated predictions and classified the content of 32 publications; column classes follow by rule from coded relations between predictions, not from a coder's judgement. The result is a concentration: only 21 of 120 cells carry a stated prediction; 6 of 12 accounts state one about the cortical signature, the one contested column — of 54 cross-theory pairs of stated predictions there, 2 are discriminating. Elsewhere commitment is sparse: the number and the incompatibility of competing action policies and visceral modulation of access have one explicit occupant each; interoceptive content three and autonomic conflict two, none in conflict; cortical-autonomic coherence none. The empirical base mirrors this: 16 of 23 theory-addressed publications manipulated neutral visual content, 0 of 23 valenced content. The next adversarial test needs a condition on a quantity the theories have committed to and never measured. We specify one: incompatibility of licensed policies at fixed count under no-report access; one confirmatory contrast tests a one-step derivation (same condition, different outcome) from a stated prediction also tested on its own variable.

Keywords: consciousness; adversarial collaboration; conscious access; theory audit; stated predictions; action-policy competition; interoception; column typology

1. Introduction

For thirty years the standard complaint about consciousness science was that its theories were not tested. The complaint has been answered, and the answer was worse. The Cogitate consortium ran a preregistered adversarial test of global neuronal workspace theory (GNWT) and integrated informa-

tion theory (IIT) across fMRI, MEG and intracranial EEG in 256 participants, with pass and fail criteria agreed in advance by proponents of both theories and by a theory-neutral consortium [1], and reported that neither central prediction survived intact [2] — GNWT’s principal failure, the absent prefrontal ignition at stimulus offset, being a null not attributable to insensitivity, since the same prefrontal sites showed strong onset responses to the same stimuli. The consortium declined a verdict, adopting a Lakatosian soft-falsification frame, and GNWT proponents have published their own reading of the same data [3].

The usual lesson drawn is about the theories: too vague, too unfalsifiable, too well defended by auxiliary assumptions. We draw a different one, and not the one we drew in the version of this review first submitted, which held that the two theories had been tested on a prediction they share. That was wrong: the consortium preregistered three predictions on which GNWT and IIT diverge, and each theory passed at least one criterion and failed at least one [1,2] (Section 6.4). The test discriminated — on a single quantity, the cortical signature of conscious content — and the consortium’s remark that its challenges extend to any theory sharing the predictions tested [2] is a remark about how many theories have staked their commitments on that one quantity.

That is the thesis of this review: the stated commitments of theories of consciousness concentrate on a few measurable quantities, and most of what the theories could be tested on is unmeasured — not because the measurements are unavailable but because no theory has said what it expects. Since “unmeasured” covers cases that license different inferences, each measurable quantity — each *column* of the audit in Section 7 — is classified by what the coded theories have stated over it, as *contested*, *occupied but not contested*, *single-occupant*, *thin* or *unoccupied*; the class is computed from the coded cells, not assigned by a reader, and Section 7.3 defines the classes beside Table 1. Only a contested column offers a result that favours one occupant over another, and silence is not a contrary prediction: an experiment on a single-occupant or thin column adjudicates between none of the silent theories, and its informative outcome is the null. Which quantity falls into which class is the question Section 7 answers; Section 9 draws the design consequence.

Locating such a quantity requires knowing what the theories actually predict, as opposed to what they are said to predict in review-article summaries; that is this review’s contribution. The purposive audit of Section 7 coded the stated predictions of 12 accounts — 10 trial-level theories of conscious access and 2 origin-level accounts reported apart — against 10 measurable quantities (120 cells), entered each stated prediction in a register (27 stated by the theories, 6 derived by us and labelled as such) and every cross-theory pair of them in a pair register, and classified separately the content whose access was manipulated in 32 publications (59 experiment-level rows) these theories are tested by or built on. It is not a systematic review, its counts are of coded accounts rather than estimates of a population, and its codes are a single coder’s until the scheduled blind second coding is complete. It is evidence for a claim otherwise easy to assert and hard to check, and one of its results contradicts our own submitted version: the column we had taken to be contested is, once the predictions are written down beside each other, occupied by accounts answering different questions.

Three conventions govern what follows. This is an evaluative review, not a census: every section closes with a judgement about which position the evidence supports, and where a widely repeated claim is weaker than its reputation we say so. We distinguish throughout between claims about *access* — content becoming available to many consumers, and claims about *phenomenal experience*. We use

one definition of “conscious access” throughout, the one in general use — content is accessed when it is reportable and available to multiple downstream cognitive systems — with no further clause built into it. That access measurably alters organism state is a hypothesis we advance, the question column M5 asks, and it is kept outside the definition so that the coding of M5 cannot be definitional: each M5 cell asks whether *that theory* states a prediction over organism-wide cortical–autonomic coherence. What the hypothesis affects is how much a thin column *matters*, an argument Section 7.4 makes openly, where it also states why the heartbeat-evoked-response literature is coded as compatible with M5 rather than as a stated prediction over it. Where a formulation carrying neither commitment is needed we use *global availability*, in the descriptive sense of content being available to many consumers rather than in the theoretical sense the term has inside global workspace theory [4].

The review proceeds from conceptual bookkeeping to measurement: Section 2 asks which distinctions the field has earned; Sections 3 to 5 evaluate the three literatures supplying the raw material for any account of why an organism would need conscious access at all — biological self-maintenance and basal cognition, action arbitration, and valence as organism-wide reconfiguration; Section 6 maps the disputes between theories and what the adversarial test did and did not decide; Section 7 presents the audit and reports the class of each coded quantity as the coding returned it; Section 8 examines three unmeasured dissociations; Section 9 states the minimal measurement that follows from the classification, and says exactly which stated prediction it tests. A theory making predictions about the single-occupant and thin coordinates exists and is presented separately; we name it once, at the end, and do not argue for it here.

2. Earned and unearned distinctions

Any claim about consciousness is a claim about one of at least seven things, and most disputes here are about which one. The coding in Section 7 depends on holding them apart: unconscious processing; conscious content; access; online metarepresentation of one’s own processing; control; post-hoc reconstruction; the putative bearer of phenomenal states. These are not degrees of one variable but dissociate pairwise.

The oldest distinction is also the least stable. Block’s separation of phenomenal from access consciousness [5] made the inference from “globally available” to “felt” a step requiring argument, later given empirical teeth by the overflow claim that experience contains more than the system can report [6,7]. The programme that grew from it — isolating the correlates of conscious content from its prerequisites and consequences [8], and distinguishing conscious state from content [9] — is a genuine achievement, and the no-report literature of Section 6 its most valuable product.

What has changed is confidence in the two-kinds framing. An argument from inside the experimental camp holds that dividing consciousness into two *kinds* converts a graded multi-component phenomenon into two entities, one inaccessible to measurement by construction [10]. That is a conceptual paper, not a result, but it converges with an independent movement toward feature profiles: multidimensional assessment of animal sentience replacing a single scale [11], indicator lists for invertebrates [12], a framework treating consciousness as a profile of separately assessable capacities [13].

The field has earned the process/content/access distinctions and has *not* earned the phenomenal/access dichotomy in its strong form, and the two are usually run together to the detriment of both. The first set is earned because each distinction is independently measurable and has produced dissociations that survived attempts to explain them away: integration without awareness [14], frontal correlates tracking report rather than the percept [15], conscious state varying independently of content [9]. The dichotomy is not, because its central claim — a second kind of consciousness whose presence comes apart from access — is what no measurement can establish; overflow arguments turn on the interpretation of partial report, the opposing position denies the separation is coherent [16], an intermediate view treats access as partial and hierarchical rather than all-or-none [17].

The cost of conflating these levels shows up in how theories get evaluated. A theory specifying when content becomes globally available answers an access question, and its success says nothing directly about phenomenality; but with the dichotomy in the background such theories are read as accounts of phenomenal experience, criticised for failing a task they never took on. The converse error is as common: theories staking a claim on phenomenality are credited with the access evidence. Feature profiles help by forcing each claim to name its own observable, which is what makes the coding of Section 7 possible.

3. From regulation to a minimal bearer

The literature on biological self-maintenance rigorously accounts for what makes a boundary an organism's own; what it does not give, and largely does not claim to, is any route from organisation to a felt aspect.

Autopoietic theory describes a living system as a network of processes producing its own boundary, treating this closure as sufficient for “cognition” [18]. Di Paolo's refinement is decisive: bare autopoiesis is insufficient even for agency, since a system that either maintains its organisation or disintegrates has no gradations of better and worse and so no norms; what is needed is adaptivity, regulating one's own relation to viability boundaries [19]. **Neither text posits a felt aspect.**

The corresponding corpus is basal cognition [20]. A parallel line proposes the skin-brain thesis: early nervous systems arose not as input–output machinery but to coordinate whole-body contraction [21]. That places single-body coordination at the origin of nervous systems — Section 4's starting point.

The split is over whether regulation entails experience. A moderate position keeps the functional vocabulary: cellular and bioelectric networks as problem-solvers in morphospace, the self's computational boundary set by the scale of coordination [22]. A radical position takes the step the moderate one declines, deriving sentience from cellular organisation and locating the first minds in single cells [23] — a redefinition rather than a proposal, adding no mechanism.

Three ladders locate the threshold differently. Feinberg and Mallatt place it in the Cambrian, tied to isomorphic neural representations [24]. Ginsburg and Jablonka propose unlimited associative learning as a *marker*: a detectable sign not a definition [25]. The third is homeostatic: feeling derived from regulation of the internal milieu [26], affect localised subcortically [27], a free-energy formalisation [28].

The anti-corticocentric line pushes the shared claim — the threshold is not the presence or absence of a cortex — lower than most expect. Merker places the core of the conscious state in upper brainstem and mesodiencephalon, where body schema, action space and motivational ranking integrate, cortex supplying content rather than consciousness [29]. This consolidated into the New York Declaration on Animal Consciousness [30].

The transition from organisation to experience is warranted nowhere here and claimed seriously nowhere except in the radical cellular position — a strength rather than an embarrassment, since the basal cognition programme is valuable precisely because it stops short. Where the inference is warranted is narrower: from organisation to a *control problem* whose solution needs something the organisation does not contain. Coordinating many specialised subsystems around one body is such a problem, solved by nervous systems from their origin [21], and it is where the ladders converge. That convergence is the difficulty: because three independent programmes reach it, “integration around a single body matters” is not an available contribution. The unoccupied position is narrower — the *number and heterogeneity of conflicting domains* integrated is a measurable variable — and no ladder contains that axis, though transitions in cognitive evolution are described as changes in control regime [31].

4. Arbitration as a function

If conscious access exists because incompatible demands from many specialised systems must be reduced to one action of one body, the mechanisms performing that reduction are where to look. Single-body arbitration has been solved unconsciously three times over, in three formalisms, none requiring access.

The first solution is anatomical. The basal ganglia were proposed as the vertebrate solution to the selection problem: many command systems compete for one body’s skeletal motor apparatus, and a centralised device is needed to resolve the conflict without being one of the competitors [32]. Formalisation showed striatal, subthalamic and pallidal microanatomy implementing an off-centre on-surround scheme [33]. This arbiter operates entirely without access [34].

The second solution is computational. Bounded accumulation of evidence to a threshold reproduces reaction-time distributions and accuracy jointly, parietal activity tracking the integral of evidence for an option [35].

The third solution is architectural. The affordance competition hypothesis holds that the visual system specifies several available actions at once while fronto-parietal circuits maintain them as mutually inhibiting population representations biased by value [36]. With two permissible movement directions, monkey dorsal premotor cortex carries two directional activity peaks simultaneously, and only after a cue is one suppressed and the other amplified to threshold [37].

Supramodular interaction theory (SIT) holds that phenomenal states exist to permit interaction among supramodular response systems — agentic, encapsulated structures defined by concerns rather than input modality — which compete for the same skeletomotor output and which, without phenomenal states, could not influence action *jointly* [38]. The passive frame formulation narrows the claim to the somatic nervous system, the mechanism passive and low-level [39].

SIT's empirical base is of three unequal kinds, stronger than its critics allow. The first is taxonomic: introspectively penetrable conflicts (pain for gain, breath holding, delay of gratification) are separated from impenetrable ones (pupillary reflex, ventriloquism and McGurk effects), the dividing line running through a skeletomotor plan. The second is experimental: participants trained to introspect rated the urge to err while sustaining compatible versus incompatible skeletomotor intentions, incompatible intentions producing stronger subjective effects while a smooth-muscle control produced none [40]. That control carries the load: without it the theory would predict a subjective effect from any antagonism. The measure in both conditions is a rating of experienced conflict. It shows that incompatible skeletomotor intentions are accompanied by a reportable state; it does not show that the arbitration required that state, nor that the state is conscious access to anything beyond the conflict itself. Arbitration is demonstrated by the outcome of the competition, access by a separate measure, and the experiment supplies the first and reports the second. Section 9 accordingly treats the step from experienced conflict to access as a derivation of ours, labelled as such, rather than as a claim of the theory. The third support is neuroimaging and markedly weaker: subjective conflict was associated with left postcentral, that is somatosensory rather than executive, cortex [41]. The sharpest question is Clark's: did voluntary skeletomotor action arise for control of the external environment or the internal one, with voluntary regulation of internal-state conflicts as a source of visceral awareness [42].

Our judgement has two parts. First, the three unconscious solutions establish decisively that policy competition is *insufficient* for access: competition is present in all three and access is absent. They establish nothing about its necessity. Necessity would require showing that access does not arise without policy competition, and nothing reviewed here bears on that; and the existence of arbitration, however well demonstrated, establishes nothing about consciousness in either direction, since the three unconscious arbiters show that arbitration proceeds without it and no experiment reviewed here shows that access changes when arbitration does. §9 accordingly tests the one stated prediction over the arbitration situation — the supramodular threshold — together with its extension to access, which no theory has stated. Any theory claiming that access exists *for* arbitration therefore owes an account of what distinguishes the arbitration requiring access from the arbitration that does not; pointing at the existence of conflict is not that account.

Second, SIT does supply such an account, and it is *taxonomic* rather than computational: the conflict either sits at the action-selection stage and involves skeletal musculature, or it does not. This is the strongest available answer, and it pays a price: the classifying feature says where the conflict is located but not what makes it unresolvable without access, so the explanation cannot transfer without re-running the taxonomy. A related cost attaches to the passive frame formulation: the conscious field presents contents but does not select, and operates whether or not conflict is present [43]. But a frame passive with respect to selection has no variable that can differ between arbitration problems, as a discriminating prediction requires. SIT buys defensibility with a commitment hard to extend, and the cell where an extension would be tested — non-skeletal conflict — is where it states a null: conflict within a smooth-muscle effector, with no skeletomotor plan, produces no experienced conflict [40]. The null is an exclusion of autonomic conflict from the function of the conscious field, not a denial that bodily states are felt — the same programme names tooth pain and urges among the most basic conscious contents, and its authors declined an invitation to extend the

theory to the autonomic nervous system rather than deny that autonomic events are experienced [39]. Section 7.4 records what that distinction does to the column it sits in.

5. Valence and organism-wide reconfiguration

An arbitration account needs a common scale. The argument that incommensurable motives must become a single quantity for behaviour to be selectable, with pleasure as that currency, was made decades ago [44], and neuroeconomics gave it a neural reading as a common currency of subjective value [45]. The problem is posed *independently of theories of consciousness*, so a theory of access saying something about the scale makes a claim none of its competitors makes.

The interoceptive literature has meanwhile moved the body from monitored object to control loop, from afferent representation of homeostatic state in insula [46]. The decisive step was redescribing it as *inference* rather than readout [47], regulation proceeding by autonomic, endocrine and behavioural action bringing the body to the predicted state [48].

Organism-wide scope comes from allostasis rather than interoception as such: set points change in advance on anticipated need [49]. Functional connectivity identified a large-scale network named allostatic-interoceptive [50]. Two caveats are load-bearing: that network came from resting-state and meta-analytic methods, not causal manipulation, so it is architectural plausibility rather than demonstrated mechanism; and a recent treatment separates interoception as *modelling* the body from allostasis as *controlling* it [51].

The thesis that an emotion brings many subsystems into a compatible regime is well supported and already owned three times over. Component process theory defines emotion as synchronisation of organismic subsystems [52]. The evolutionary tradition describes emotions as operating modes tuning many parameters for a situation type [53]. Emotion psychology adds control precedence: emotion seizes control and reconfigures action readiness rather than adding content [54]; the constructionist line says the same in allostatic vocabulary [55].

The empirical base is weaker than the theoretical statements suggest. Neuroimaging meta-analysis found no category-specific brain signatures for discrete emotions [56]; meta-analysis of autonomic patterns found the same for the periphery [57]. The two nulls do not show that global reconfiguration occurs; they show that if it occurs it takes no reproducible configuration per category. The evidence that it occurs is the evidence of the previous paragraph — synchronisation of subsystems, control precedence, operating modes — and that evidence is theoretical statement and behavioural observation rather than a measured organism-wide state. What the nulls do settle is the unit of analysis: not a category with a signature but a population of states sharing a functional task [58]. Formal models deriving valence from regulatory dynamics rather than positing it — the rate of change of prediction error [59] — are proposals rather than results, but the closest formal analogue available.

Two refinements are non-optional. First, valence is not one parameter: hedonic “liking” and motivational “wanting” dissociate neurochemically and anatomically [60], so a one-dimensional good/bad axis coarsens what is measured. Second, controlling and experienced valence must be separated. The two-system framework holds that subcortical survival circuits produce defensive

responses and bodily mobilisation while experienced fear is a separate cortical product [61]; in the stronger version, experienced emotion requires higher-order representation of the state as one's own [62], resting on the priority of metarepresentation in conscious access [63].

The LeDoux and Brown critique is correct as a constraint and overextended as a conclusion. It is correct that an integrated valent configuration can exist without being experienced, so no account may treat organism-wide valent integration as sufficient for a putative bearer of phenomenal states. It is overextended in inferring that experienced valence requires the higher-order architecture proposed: the step from “integration is not sufficient” to “metarepresentation is what is missing” is not supported by the dissociation data — the affect-first side supplies a rival subcortical mechanism [27,28]. The pain literature is the best model for keeping three levels apart. Nociception is phylogenetically broad and uncontroversial [64]; operational criteria for sentience rest not on it but on features requiring integration — motivational priority shifts, trade-offs, self-administered analgesia [65].

The judgement is that the middle level is real, measurable and under-theorised. Global defensive or appetitive reconfiguration is measurable through motivational priority and trade-off behaviour rather than report, **though it has not yet been measured as a joint cortical–autonomic state on a trial, which is the sense in which Section 7 calls the corresponding column thin**, and is where the arbitration and valence literatures meet. What no one has done is make the *structure* of that configuration a function of the *structure* of the conflict resolved: allostatic and constructionist accounts derive affect from regulation and prediction error, neuroeconomics assumes the scale, SIT has no affective variable. The proposition that the number and heterogeneity of conflicting domains sets the required dimensionality of valent integration appears unclaimed — and unlike most such propositions, it is measurable.

6. Theories and their empirical tests

The theories are easier to describe as a list of disagreements than of schools. Almost everyone agrees that conscious content is connected to broad availability; the disagreements are about what makes it conscious — network topology, causal structure, second-order representation, a predictive model of the body. We organise the map by dispute, where the empirical blows of the last five years landed, most striking several theories at once.

6.1 Where content lives

The localisation dispute — posterior versus prefrontal — was the most productive, because both sides stated mutually exclusive predictions. The workspace line ties access to late global ignition in a network privileging prefrontal cortex [66,67], its strongest evidence nonlinearity: a second attentional-blink target's fate is bimodal rather than graded [68], and the masking threshold coincides with a sharp inflection in late components [69]. Notably, GNWT's rationale for broadcast is flexibility rather than arbitration — an answer to the frame problem [70], a serial router for composite mental programmes [71]. IIT proceeds from phenomenological axioms to maximally irreducible integrated information, placing the hot zone in posterior temporo-parieto-occipital cortex [72,73], experience identified with the distinctions and relations unfolded from a substrate [74]. Recurrent processing theory holds instead that local recurrent activity in sensory cortex makes content conscious, entailing phenomenal content without access [75,76]. The dispute was long

reproduced positionally [77], with the well-founded suspicion that outcomes depend on method and on who runs the experiment [78].

6.2 Who predicts phenomenality

Few theories predict phenomenality rather than access, and the price is metaphysical load or denial of the explanandum. The claim concerns predictive content, not the number of frameworks that address phenomenality. Kuhn's [79] landscape arrays theories of consciousness by ontological commitment — materialist positions in several subcategories, non-reductive physicalism, quantum theories, integrated information, panpsychisms, monisms, dualisms and idealisms — each self-described by its adherents, with critique kept to clarification and no attempt to adjudicate among them; its organising axis is what a theory says consciousness *is*, not which observable it stakes. The inclusion rule of Section 7 cuts across that landscape: a framework enters Table 1 only if it states a prediction over a measurable quantity, so whole regions of Kuhn's map contribute no row — not because they are mistaken but because they answer a different question. Poznanski [80], for example, proposes a constitutive account on which conscious actuality is the intrinsic realisation of a transscale ontophysical continuity; the framework addresses phenomenality at the ontological level and, as presented, states no measurable prediction, and so falls outside the inclusion rule of Section 7. Higher-order theories predict structurally: a second-order representation constitutes being conscious, so metacognitive sensitivity should dissociate from intact first-order processing [63,81]; the most operationalised variant nests a one-dimensional awareness state above a high-dimensional perceptual space [82], with consequences for confusing perception and imagination [83]. Attention schema theory has a system attribute awareness to itself because it models its own attention [84,85], and has been offered as a reconciliation of AST, GNW, HOT and illusionism [86]. Predictive processing explains the structure of experience rather than its presence — perception as controlled hypothesis [87], interoceptive inference as the basis of emotion and bodily selfhood [47,88] — and claims only a basis for correlate-hunting [89]. The poles bound the space: the hard problem as non-derivability of experience from function [90], the meta-problem with illusionism as its answer [91], and the more modest working stake of making the hard problem operationally easier through accumulated structural correspondences [92].

The methodological threat is logical rather than empirical. The unfolding argument shows that a theory defining consciousness by causal structure assigns different values of its measure to a recurrent network and to a functionally equivalent unfolded one that behaves identically, reports included, so the theory is either false or behaviourally indistinguishable from its opposite [93]; responses rejected its behaviourism [94] or located a hidden premise about brain states [95]. The problem generalises: falsification requires independent access to phenomenality, without which every theory is protected from refutation [96] — hence criteria for empirical theories [97], a Lakatosian accounting scheme [98], and the separation of a testable weak core from a loaded strong one [99]. The 2023–2025 controversy over whether IIT is pseudoscientific continues the same dispute, and we report it without taking a side [100,101].

6.3 Access without report

Dependence on report biases toward frontal theories, and the key result is direct: in binocular rivalry, frontal activity tracks introspection and the act of reporting rather than the percept [15]. The

programmatic statement [102] and the isolation of perceptual from report-related correlates [103] extend to a report-independent bifurcation signature [104], large-scale networks in a no-report visual paradigm [105] and audition [106]. Calibration is needed both ways: report reliably adds frontal correlates, but null frontal results do not themselves bury prefrontal theories [107], and no-report designs have their own defect — without report one does not know what the participant experienced [108].

6.4 What Cogitate decided, and what it left open

Exactness matters, because the loose version circulates widely. The consortium preregistered five points of divergence between GNWT and IIT and tested three of them as critical: the locus of decodable conscious content (prefrontal and parietal cortex in tandem with high-level sensory cortex, against the posterior hot zone), the temporal profile of its neural correlate (a transient ignition at stimulus onset and again at offset, against activity sustained for the duration of the percept), and the connectivity that carries it (fronto-posterior synchronisation, against synchronisation within posterior cortex) [1,2]. Each theory passed at least one criterion and failed at least one. For IIT the most direct challenge was the absence of sustained within-posterior synchronisation; its content-maintenance prediction passed the pre-set criteria — sustained responses reflecting stimulus duration appeared in occipital and lateral temporal cortex — but sustained representation of orientation did not, although orientation decoded reliably in all three modalities. For GNWT the principal failure was the absence of ignition at stimulus offset, with a secondary challenge on content: category decoded in prefrontal cortex independently of task, identity not at all, orientation only in MEG — raising the question whether prefrontal cortex broadcasts all conscious content or only an abstract part [2].

The adversarial test therefore did what an adversarial test is for. Predictions were fixed in advance, criteria were agreed by opponents, results were reported against them rather than reinterpreted afterwards, and the three preregistered predictions stand in two relations to each other in our pair register. On the temporal profile — sustained content-specific activity for IIT, an offset-locked ignition for GNWT — the two predictions name one observable on which each excludes the other's value, and the test separated the theories there: sustained posterior responses appeared and offset ignition did not. On locus (P01–P04, decodability of content after onset) and connectivity (P03–P06, sustained synchronisation) the register enters the GNWT–IIT pairs as jointly compatible: each names a shared observable, but the value predicted by one does not exclude the value predicted by the other — prefrontal and posterior decoding can both hold, and so can prefrontal-posterior and within-posterior synchronisation — so on those criteria each theory failed its own test — IIT on sustained posterior synchrony, GNWT on the identity and orientation content of prefrontal decoding — without the failure favouring the other. The test therefore separated the theories on one value and let each fail its own criterion on the others; the earlier version of this review was wrong to describe the outcome as a test that could challenge many theories and separate none. What the test did not do is decide anything outside the quantity it varied. The consortium's remark that the challenges extend to theories sharing the predictions tested [2] is exact — recurrent processing theory shares the posterior locus, and higher-order theories that treat prefrontal cortex as a supplier of content share the prefrontal one — but it reaches them only on that quantity. Whether the same theories have committed themselves to anything else, and whether any two have committed themselves to opposed

values of anything else, is a question about their stated predictions, and it is answerable by writing them down side by side. Section 7 does that.

7. The measurement gap

7.1 What is coded and why

Theories of consciousness are compared, almost exclusively, on where and when conscious contents are encoded. To establish how far stated commitments extend beyond the one quantity the first adversarial test varied (Section 6.4), rather than assert it, we carried out a purposive audit: the stated predictions of 12 accounts were coded against 10 measurable quantities from primary statements and self-profiling reviews (Table 1; 120 cells); each stated prediction was entered in a register with its source, condition, outcome measure and expected result (Table S4; 27 stated and 6 derived by us), and every cross-theory pair of stated predictions in a column in a pair register with its relation; and the content whose access was manipulated in 32 publications (59 experiment-level rows) these theories are tested by or built on was classified (Table 2). *Purposive* is doing work: there was no registered search, no PRISMA flow and, at the time of writing, one completed coder (Section 7.2), and the tallies are counts of coded accounts, never fractions of a population of theories.

The theory set was assembled under one inclusion rule — an account qualifies if it has at least one preregistered or explicitly stated empirical prediction about conscious access in the current literature, which excludes purely metaphysical and purely descriptive frameworks — and one coverage aim: to span the families against which the field’s empirical corpus is organised — the four leading theories of the ConTraSt database [78], GNWT, IIT, recurrent processing theory and higher-order theory, with the attention schema theory — and every family we could identify that states a prediction over one of the quantities M1–M5 the corpus does not track: the supramodular family (supramodular interaction theory and its passive-frame successor), the interoceptive-inference family (predictive processing and the beast-machine account, and the neural subjective frame programme), and the two origin-level accounts, unlimited associative learning and the Feinberg–Mallatt scheme. The rule does not pick out these rows and no others: dendritic integration theory (apical–basal coupling in layer-5 pyramidal neurons, tested under anaesthesia [109,110]), the temporo-spatial theory of consciousness (pre-stimulus temporo-spatial alignment [111,112]) and active inference as an account distinct from predictive processing (a formal workspace model [113]) meet it and were not coded. We name them so the omission is visible and a second coder can add them.

Two rows of the earlier table are reported apart: unlimited associative learning [114,115] and the Feinberg–Mallatt scheme [24] answer when, and in which lineages, consciousness appeared, not what makes a content accessible on a trial, so their silence on the cortical signature records a difference of subject matter, not a hole in a predictive apparatus, and they form their own block in Table 1 with their own tally. Two pairs of rows previously merged are now coded from their own statements: supramodular interaction and passive frame theory, because Section 4 argues that the passive-frame formulation gives up the variable a discriminating prediction needs; and higher-order thought [63,81] and higher-order state space theory, the latter offered first as a model of *reports* of awareness, treating higher-order inference as necessary but possibly not sufficient, and predicting

prefrontal involvement in decisions about stimulus absence [82]. Every cell of the two higher-order rows is provisional until the second coding confirms the split.

The quantities were not chosen for neutrality, and claiming otherwise would be indefensible: three of them (M1, M2, M5) are coordinates of the theory named at the end of this paper. The selection principle was different and is stated here so it can be argued with. Each quantity is a variable that the literatures reviewed in §§3–5 already treat as central to biological regulation, action arbitration and allostasis — not a variable invented for this survey. The test is then whether any theory of *consciousness* predicts over variables that the surrounding biology treats as fundamental. That a coordinate is also an AFT predictor is a consequence of AFT having been built from those same literatures, not evidence that the column set was rigged; but a column set chosen by the first author cannot certify its own independence, and the emptiness of a column relative to it is not evidence of anything until the column set has been checked against one the first author did not choose. Section 7.5 reports the first such check. The ten quantities: three concern the arbitration situation (M1 the number of simultaneously competing action policies, M2 their degree of incompatibility, M5 organism-wide coherence across cortical and autonomic channels); four the content (M3 valence, and the three questions into which the earlier single column “autonomic involvement” has been split because the predictions coded under it answered three different questions — M4a whether interoceptive signals can themselves become the accessed content, M4b whether the state or phase of an autonomic signal modulates access to *exteroceptive* content, M4c whether a conflict resolved wholly by autonomic effectors, with no skeletomotor plan, involves conscious experience); and three the format and signature of access (M6 dimensionality of phenomenal space, M7 precision of metarepresentation, M8 cortical signature). The split of M4 is the largest change since submission, and Section 7.4 reports what it did to the column the earlier version called contested.

7.2 Coding procedure

Each cell carries one of five codes defined in the deposited codebook (v2, frozen at version and hash). EXPLICIT: a sentence or preregistered prediction can be quoted that names the quantity and says what should happen to access when it varies. INTERPRETED: the prediction follows from the stated mechanism in one nameable step but is not stated for this quantity (the earlier version’s IMPLICIT). NOT_LOCATED: no statement bearing on the quantity was found in the examined sources; it is not a prediction of no effect and is never counted as one. NOT_APPLICABLE: the theory places the quantity outside its scope, and the scope statement is quoted. UNRESOLVED: sources of the same account conflict; the cell goes to adjudication. Every EXPLICIT and INTERPRETED cell carries a polarity — positive, null or negative — and a null is a prediction: a stated claim that the quantity makes no difference, or its exclusion from the theory’s function, is coded EXPLICIT with null polarity and reported apart from the positives. Each coded cell is anchored to a source, a locus, an evidence status (full text or abstract only) and a coder confidence from 1 to 3. The INTERPRETED/NOT_LOCATED boundary carries the analytic weight and was applied conservatively, so the count of NOT_LOCATED cells is a lower bound on the field’s silence.

The table has three states. The submitted version had 72 cells — nine accounts by eight quantities — coded NO 38, IMPLICIT 17, YES 16, YES (negative) 1. The first revision draft (deposit v1.3) had 88 cells, eleven accounts by eight quantities, after the neural subjective frame was added and the

supramodular and passive frame theories were separated: NO 45, IMPLICIT 24, YES 17, YES (negative) 2. The present table has 120 cells, twelve accounts by ten quantities, after the higher-order thought and higher-order state space theories were separated and the autonomic column M4 was split into M4a–M4c. The 88 codes of the second state were mapped, not recoded — YES to EXPLICIT positive, YES (negative) to EXPLICIT null, IMPLICIT to INTERPRETED, NO to NOT_LOCATED or, where a scope exclusion is stated, NOT_APPLICABLE — and carried as 96 legacy entries, because the merged higher-order row's eight codes were copied to both of its successors (84 in the cell-level file: NO 44, IMPLICIT 23, YES 17; the 12 codes of the superseded M4 column in the long-form file: NO 4, IMPLICIT 3, YES 3, YES (negative) 2). The mapping was applied with two exceptions on the split higher-order rows (HOT × M6 and HOSS × M3, entered NOT_LOCATED and flagged because the legacy code rested on the other account's source); the legacy code stands beside the current one in the deposited table, and the 12 superseded M4 codes are replaced by the 36 cells of M4a–M4c, coded afresh, and enter no count. The result: 22 EXPLICIT cells (20 positive, 2 stated null), 29 INTERPRETED (21 positive, 8 stated null), 67 NOT_LOCATED, 2 NOT_APPLICABLE and 0 UNRESOLVED. A cell is provisional when, and only when, its code rests on an abstract rather than the full text — 2 cells at present, whose sources (Brown, Lau and LeDoux 2019; Clark 2013) have not been obtained; the other 55 were read in full on 25 September 2026 and 6 codes changed (Section 7.7) — and 24 cells are flagged for adjudication with a written reason, the first of them predictive processing × M4c, where the mechanism is stated but the condition is not named. Five cells were re-coded by the first coder on 24 September 2026 after a second pass and carry the previous code beside the current one in Table S1; a full-text pass over the flagged cells on the same date reached 3 of 12 sources; 21 of the 28 sources under the remaining abstract-only EXPLICIT and INTERPRETED cells have an open-access copy, and their reading is scheduled before the codebook freeze (Section 7.7).

The first coding was carried out by the first author. A blind second coding by the second author is scheduled against the frozen codebook (version 2.3; the hash of the file above its footer is printed in the footer and can be checked with one shell command) and blank forms, without access to the first author's codes or justifications, on all 120 cells of Table S1 and all rows of Table S2; disagreements will be adjudicated by discussion and the pre-adjudication agreement reported. The second author has read the submitted manuscript, including its condensed Table 1; the second coding has not begun. Blindness is therefore complete for the per-cell justifications and for the cells added in revision (the M4a–M4c columns and the passive frame and neural subjective frame rows) and partial for the cells whose condensed codes appeared in the submitted version (70 of the 120 cells carry that label, 50 were added in revision); agreement is reported for the two subsets separately as well as overall. Separately from the second author's coding, and not in its place, an independent AI coding of all 120 cells against the same frozen codebook (v2.4; re-frozen as v2.5 with a completed source list before the second author's coding) was obtained on 25 September 2026 from a session that had not seen the first coder's codes, justifications or the revised manuscript; it is deposited with its own limitations stated by that session (it drew on 19 of the 49 primary sources listed in the codebook, 18 in full text and one from its abstract, and claims no experimentally secured blindness). Against the first coder it agrees on 88 of 120 codes ($\kappa = 0.55$, 95 % bootstrap interval over theories 0.44–0.66), on 85 of 120 code-and-polarity pairs, and on 118 of 120 cells at the binary level of a statement located or not ($\kappa = 0.67$); the 35 disagreements are split evenly in direction (15 where the AI coding is stronger, 14 where it is weaker, 6 on polarity or status) and enter the adjudication as a third reading. It is reported here

as a check on the coding's reproducibility from the written rules, not as the second coding the Reviewers asked for. The distinguishability judgements in Table S4 — which pairs of stated predictions differ in sign, magnitude, locus, timing, distribution or condition, and at what confidence — and the relations in the pair register (63 pairs: 54 in M8, 3 each in M3 and M4a, 1 each in M6, M7 and M4c) were made by the first coder alone; the second coder receives Table S4 and the register only after returning Table S1 and codes the relations independently. [κ = ..., raw agreement ... of 120 cells overall; κ = ... for the submitted-version cells; κ = ... for the cells added in revision; ... disagreements adjudicated by discussion; S2 κ = ... — to be completed when the second coding is finished] The tables, protocol, codebook, inclusion criteria and scripts that produce every count in this section are listed under Supplementary Materials.

7.3 Table 1: theories by measurable quantity

Account	M1 N policies	M2 Incompat.	M3 Valence	M4a Intero. content	M4b Visceral modul.	M4c Autonomic conflict	M5 Org-wide coh.	M6 Dim. phen.	M7 Metarep. precision	M8 Cortical signature
<i>Trial-level accounts (n = 10)</i>										
GNWT	·†	·	<i>i</i> ∅	·	·	·	·	·	<i>i</i> †	E
IIT (3.0/4.0)	·	·	·†	·	·	·	·	E	·	E
Recurrent processing (RPT)	·	·	·	·	·	·	·†	·	<i>i</i> ∅†	E
Higher-order thought (HOT)	·	·	<i>i</i> ∅	<i>i</i>	·†	<i>i</i>	·	·†	E	E
Higher-order state space (HOSS)	·	·	·†	·†	·†	·†	·	E	E	E
Attention schema (AST)	·	·	·	·	·	·	·	<i>i</i>	<i>i</i>	E†
Predictive processing / beast machine (PP)	·	·†	E	E	<i>i</i> †	<i>i</i> †	<i>i</i>	<i>i</i>	<i>i</i> †	<i>i</i>
Supramodular interaction theory (SIT)	E	E	<i>i</i> ∅	E	<i>i</i> ∅†	E∅	·	·†	·	<i>i</i>
Passive frame theory (PFT)	<i>i</i> ∅	<i>i</i> ∅†	<i>i</i> ∅	<i>i</i>	·	E∅	·	·	·†	n/a
Neural subjective frame (NSF)	·	·	<i>i</i>	<i>i</i> ∅†	E	·	<i>i</i>	<i>i</i> †	·	<i>i</i>
Tally, trial-level (n = 10)	1 E / 1 <i>i</i> ∅ / 8 ·	1 E / 1 <i>i</i> ∅ / 8 ·	1 E / 1 <i>i</i> / 4 <i>i</i> ∅ / 4 ·	2 E / 2 <i>i</i> / 1 <i>i</i> ∅ / 5 ·	1 E / 1 <i>i</i> / 1 <i>i</i> ∅ / 7 ·	0 E / 2 E∅ / 2 <i>i</i> / 6 ·	0 E / 2 <i>i</i> / 8 ·	2 E / 3 <i>i</i> / 5 ·	2 E / 3 <i>i</i> / 1 <i>i</i> ∅ / 4 ·	6 E / 3 <i>i</i> / 0 · / 1 n/a
<i>Origin-level accounts (n = 2)</i>										
Unlimited associative learning (UAL)	·	·	E	<i>i</i>	·	·	<i>i</i>	·	·	n/a
Feinberg & Mallatt (FM)	·	·	E	E†	·	·†	<i>i</i>	·	·	<i>i</i>
Tally, origin-level (n = 2)	0 E / 2 ·	0 E / 2 ·	2 E / 0 ·	1 E / 1 <i>i</i> / 0 ·	0 E / 2 ·	0 E / 2 ·	0 E / 2 <i>i</i> / 0 ·	0 E / 2 ·	0 E / 2 ·	0 E / 1 <i>i</i> / 0 · / 1 n/a
Tally, all rows (n = 12)	1 E / 1 <i>i</i> ∅ / 10 ·	1 E / 1 <i>i</i> ∅ / 10 ·	3 E / 1 <i>i</i> / 4 <i>i</i> ∅ / 4 ·	3 E / 3 <i>i</i> / 1 <i>i</i> ∅ / 5 ·	1 E / 1 <i>i</i> / 1 <i>i</i> ∅ / 9 ·	0 E / 2 E∅ / 2 <i>i</i> / 8 ·	0 E / 4 <i>i</i> / 8 ·	2 E / 3 <i>i</i> / 7 ·	2 E / 3 <i>i</i> / 1 <i>i</i> ∅ / 6 ·	6 E / 4 <i>i</i> / 0 · / 2 n/a
Stated predictions in the register (Table S4)	1	1	3	3	1	2	0	2	2	12
Discriminating pairs / cross-theory pairs registered (pair register)	—	—	0 of 3	0 of 3	—	0 of 1	—	0 of 1	0 of 1	2 of 54
Derived class	single-occupant	single-occupant	occupied-not-tested	occupied-not-tested	single-occupant	occupied-not-tested	thin	occupied-not-tested	occupied-not-tested	contested

Table 1. Stated predictions of 12 coded accounts against 10 measurable quantities (120 cells). Trial-level accounts are tallied first; the two origin-level accounts form a separate block with their own tally. E = EXPLICIT, positive polarity; E∅ = EXPLICIT, null polarity (a stated prediction that the quantity makes no difference, or its exclusion from the theory's function); *i* = INTERPRETED, positive; *i*∅ = INTERPRETED, null; · = NOT_LOCATED (no statement found; not a prediction of no effect); n/a = NOT_APPLICABLE. † marks a cell flagged for adjudication (24 cells). Every cell of the M4a–M4c columns and of the HOT and HOSS rows is provisional pending the second coding. The last three rows are outputs of the typology script, not codes: stated predictions in the register, discriminating pairs among the cross-theory pairs of stated predictions entered in the pair register (a dash where the column holds fewer than two stated predictions), and the derived class. IIT (3.0/4.0) denotes integrated information theory in its 3.0 and 4.0 formulations, coded as one account because their stated predictions over these quantities coincide. Condensed from Table S1.

The derived class in the last row is not coded but computed by script from Table S1, the register and the pair register after the first coding was frozen: *contested* if at least two EXPLICIT cells and at least one cross-theory pair of stated predictions whose relation is *discriminating* or *incompatible* — a shared observable under a shared condition, with a value predicted by one and excluded by the other; *single-occupant* if exactly one EXPLICIT cell; *occupied but not contested* if at least two and no such pair; *thin* if none but at least one INTERPRETED; *unoccupied* if none. Predictions that merely differ — a locus beside a timing, one axis of judgement beside the structure of experience — are entered as *different* or *jointly compatible* and do not make a column contested. Every cross-theory pair carries its relation, the result that would favour each side and a pair-level confidence from 1 to 3, and the class is reported over all pairs and over pairs at confidence ≥ 2 .

Every count below is of the form k of n coded accounts, recomputed by script from the deposited matrix. The register holds 27 stated predictions, 12 in the M8 column and 15 in the other nine together; the spread across rows is modest, the spread across columns is not, and the columns are the result. Six further entries are derived by the authors, labelled so, and not counted in Table 1: P23 (a one-step derivation from the supramodular threshold P16: the same condition, incompatible intentions, with access rather than experienced conflict as the outcome), P24 (a feeling accompanying purely autonomic conflict, pending its proponents' view), P32 and P33 (the count threshold and its equivalence half, both estimated rather than tested in Study 1), P34 (the Study 2 valence hypothesis) and P25 (a load effect of policy count, derived from GNWT with a capacity assumption it has not made); a seventh row, P35, is not a derivation but the report-based test of P16 on its own variable that Study 1 carries alongside; each is introduced where it is used (Sections 7.4, 9 and 9.1).

7.4 Reading the columns

The classes are reported as the script returned them, including the three places where they contradict the earlier version of this review.

Contested: M8. The cortical signature of access is stated by 6 of 12 accounts — GNWT, IIT, recurrent processing theory, higher-order thought theory, higher-order state space theory and the attention schema theory — and INTERPRETED for 4 more; the two NOT_APPLICABLE cells (passive frame theory, unlimited associative learning) decline the level at which the question is posed. The register holds 12 stated predictions here, six of them the preregistered GNWT and IIT predictions of the Cogitate protocol [1,2] (Table S4 lists the rest); of the 54 cross-theory pairs in the pair register, 42 are different, 10 jointly compatible, 2 discriminating and none incompatible outright. The GNWT and IIT predictions on temporal profile (P02, P05) name one observable — content-specific activation across a 0.5, 1.0 or 1.5 s stimulus — and predict values that exclude each other, an offset-locked burst against sustained activation with no offset event; this is the pair Cogitate tested [2], and the class rests on it (pair confidence 2). The second, recurrent processing against higher-order theory on the necessity of dorsolateral prefrontal function for awareness at matched first-order performance (P08, P09), is entered at pair confidence 1 and flagged: it discriminates only if phenomenal content and metacognitive awareness are one variable. The earlier version counted 15 “distinguishable” pairs here and called the column “shared and convergent”; under the four-way relation 7 of those are different, 6 jointly compatible and 2 discriminating, and the column is contested on the strength of the two, not

the fifteen — shared at the level of the quantity, contested at the level of one value, the temporal profile.

Single-occupant: M1 and M2. The number of simultaneously competing policies (M1) and their degree of incompatibility (M2) are each stated by 1 of 12 accounts, supramodular interaction theory, whose PRISM principle makes the simultaneous arrival of incompatible inclinations at the skeleto-motor output system the trigger for conscious conflict [38,39], measured as Section 4 describes [40]. Two features of the prediction (P16) matter for Section 9: it is a threshold, not a count effect — conscious conflict arises when two incompatible inclinations are active at once, and the sources say nothing about more than two, a NOT_LOCATED about the further increase rather than a prediction that there is none [39] — and its outcome variable is experienced conflict, not access to a stimulus (P17), so the access outcome that Section 9 tests is our one-step derivation, P23 — the same condition, a different outcome — while P16 itself is tested on its own variable in the same sessions. Passive frame theory is INTERPRETED with null polarity on both columns, since the field presents its contents whether or not conflict is present [39,43]; predictive processing, INTERPRETED on M2 in the first pass, is NOT_LOCATED on a full-text reading. No other account states or entails a prediction over either quantity.

Occupied but not contested: M3 and M7. Valence (M3) is stated by predictive processing at trial level and by both origin-level accounts at lineage level, with the same sign; 5 accounts are INTERPRETED, 4 of them with a constructed null (they admit valenced content and state no valence-dependent access prediction), and the IIT cell is flagged because its statement that affect is not an essential property of experience can be read as a stated null rather than silence. Metarepresentational precision (M7) is stated by higher-order thought theory and higher-order state space theory [63,82], both as a dissociation of a metacognitive index from first-order performance, and inferred for predictive processing [116], whose sources state precision as a control parameter without linking it to the presence of experience (a cell re-coded from EXPLICIT after full-text reading, Section 7.7); the one cross-theory pair of stated predictions is jointly compatible. Both columns can corroborate or disconfirm their occupants together and cannot separate them.

Occupied but not contested, formerly called contested: M4a and M4c; single-occupant: M4b. Splitting the earlier single column into three questions dissolves the opposition the earlier version had prized — supramodular interaction theory's stated negative against an interoceptive-inference positive. Access to interoceptive content as content (M4a) is stated by predictive processing, for which visceral channels are constitutive of feeling [48], by supramodular interaction theory, for which a visceral process is experienced only in the phases that require coordination with skeleto-motor plans [38,39], and by the Feinberg–Mallatt scheme at lineage level [24] (a flagged cell, Section 7.7); 4 accounts are INTERPRETED positive and the neural subjective frame INTERPRETED null, the frame being proposed as a referential not itself experienced as content [117]. Visceral modulation of access to exteroceptive content (M4b) is stated by the neural subjective frame alone, whose anchor is that pre-stimulus heartbeat-evoked responses predict detection of a faint grating independently of measured cardiac parameters [118]; predictive processing is INTERPRETED, re-coded from EXPLICIT because the coded sentence reviews evidence rather than stating a prediction; 10 accounts are NOT_LOCATED. Conscious involvement in purely autonomic conflict (M4c) carries two stated nulls: supramodular interaction theory predicts, and has measured, that conflict within a smooth-muscle

effector produces no experienced conflict [40], and passive frame theory states the same boundary as scope, the conscious field serving the somatic nervous system [39]; both exclude autonomic conflict from the function of consciousness and neither denies that bodily states are felt (Section 4). Against these nulls stand no stated positives: the interoceptive-inference sources contain no statement about a conflict resolved by autonomic effectors alone [47,48], the prediction that unresolved interoceptive prediction error would yield a feeling is our derivation (P24), and the one passage bearing on it names predictions rather than prediction errors as the likelier vehicle of conscious feeling [48]. The positives on M4a and the nulls on M4c answer different questions; the three M4a pairs are jointly compatible (the supramodular condition — phases requiring skeletomotor coordination — is a nameable difference from the predictive-processing and lineage claims, but neither side states the other's result as excluded), no pair in either column is discriminating, and each is occupied but not contested; M4b is single-occupant. What would make M4c contested, and what a test of either column additionally requires, is stated in Section 9.2.

Thin, not empty: M5. Organism-wide coherence across cortical and autonomic channels, a joint measure of the state accompanying access, receives 0 EXPLICIT and 4 INTERPRETED cells among 12 accounts, and 8 NOT_LOCATED. M5 is the construct Section 1 proposed, and the column records that no coded theory has adopted it: four are compatible with it and none operationalises it as an organism-wide joint measure. GNWT's global availability is organism-wide in rhetoric and cortical in measurement, its dependent variables being ignition and P3b [119] — the full text of the taxonomy paper states long-distance synchrony only between cortical areas, so the earlier inference to an organism-wide measure was withdrawn and the cell is NOT_LOCATED; allostatic accounts identify a large-scale interoceptive system [50] without proposing its coherence with cortical dynamics as an access criterion. The nearest stated prediction is the neural subjective frame: cortical responses to heartbeats are held to be a precondition of conscious perception [117], and pre-stimulus heartbeat-evoked responses do predict visual [118] and somatosensory detection [120,121], as does respiratory phase [122,123], and have detected residual consciousness in post-comatose patients [124]. These measures couple cortex to one visceral channel and are, by the original authors' own control analyses, independent of the measured autonomic state [118]; they index cortical monitoring of a visceral signal rather than coherence across channels, and the programme frames the quantity as necessary but not sufficient for conscious perception [117] and as distinct from attention [125]. We code it INTERPRETED on M5 and EXPLICIT on M4b, and withdraw the earlier formulation that nobody operationalises a cortico-visceral quantity as a condition on access: somebody does, single-channel, and the column is thin rather than empty. A coder who read the heartbeat-evoked response as a cortical–autonomic coherence metric would fill 1 of 12 cells and make the column single-occupant, which is why M5 is a secondary and qualified claim in this review (Section 7.7).

Occupied but not contested: M6. Dimensionality of phenomenal space is stated by IIT, for which the structure of an experience is identical to the unfolded cause–effect structure of its substrate and therefore high-dimensional [74], and by higher-order state space theory, which nests a one-dimensional awareness state above a high-dimensional perceptual space [82]. The pair register enters the pair as *different*, the two statements addressing different variables — the geometry of what is experienced against the dimensionality of the judgement that marks a state as experienced — with no shared observable under which one value would exclude the other; the earlier version had listed it

as distinguishable at confidence 1. The column returns to contested only if adjudication rules, on the sources, that the two claims concern one variable. Three accounts are INTERPRETED, and two cells are flagged, one in each direction.

The single-occupant and thin columns intersect in a cell no study fills: access to valenced or interoceptive content under a manipulated number of incompatible policies, measured across cortical and autonomic channels jointly. Part of that cell is occupied: Morsella and colleagues established that policy incompatibility has measurable consequences for experience and that smooth-muscle conflict does not [40,126]; Desender and colleagues, that experienced conflict is a separable quantity, dissociable from conflict adaptation [127] — the closest precedent for treating M2 as a predictor of experience; Garfinkel and colleagues, that interoceptive accuracy dissociates from interoceptive awareness [116]. None has varied conflict load orthogonally to salience while measuring access with a report-independent signature, the constraint the no-report literature established as necessary [102], with objections in [108]; nor can the Cogitate dataset be reanalysed to fill the cell, since its stimuli — faces, objects, letters and false fonts under target detection — hold conflict load constant by construction [2]. A gap reducible to unavailable methods is a difficulty, not a gap, and most of what the missing measurement requires is routine: policy conflict is manipulable and its subjective consequences measurable without overt movement [40]; report-independent access measures are established in the visual [15,103] and auditory [106] modalities; calibrated valenced stimuli [128] and standardised cardiac interoceptive measures [116] are off the shelf; and multi-site preregistered adversarial designs are feasible at scale [1]. The one component that is not routine, which decides what in Section 9 is ready to run, is named in Section 9.2. The absence is explained instead by what the theories ask for — support for a theory is predictable from methodological choices alone, and most studies interpret results post hoc rather than testing a priori predictions [78] — and where no theory names conflict load or organism-wide coherence as a variable, no paradigm is built to vary it.

7.5 An independently generated column set

Section 7.1 conceded that emptiness relative to a column set of the first author's choosing is evidence of nothing until the set is checked against one the first author did not choose. The first check we could run uses the dimensional scheme of the ConTraSt database [78], assembled by people with no stake in this argument to organise 412 experiments, and codes the theory rows against its twelve dimensions (Table S3; 120 cells; the scheme, its provenance and its limits are described in Supplementary Note S6).

The result has two parts. First, along the independent column set every dimension receives at least one stated prediction; none is empty in the sense that M5 is thin in Table 1. Per row, GNWT states predictions on 11 of 12 dimensions, IIT on 10, recurrent processing theory on 7, passive frame theory on none, because it declines the implementation level; the sparse dimensions are the electrophysiological result fields (frequency findings one stated prediction, temporal findings three). The theories are therefore not uniformly underspecified, and the thinness of M5 is not an artefact of general vagueness: the same accounts that are silent on M5 speak on most of the axes the field's own inventory is organised on. Second, none of M1, M2 or M5 corresponds to a ConTraSt field, the check covering every field name and value in the live schema, whereas M7 and M8 are directly represented

and M3, M4a–M4c and M6 partly recoverable through free-text labels. The quantities this review calls unmeasured are also unrecorded in the largest theory-neutral inventory of the field.

The check establishes that the Table 1 result for M1, M2 and M5 is not the rediscovery of an existing axis under another name. It does not establish that the quantities are absent from experiments, since ConTraSt records what authors reported, and it checks occupancy, not disagreement. Its own limits — a scheme reconstructed from the database’s methods description, ten rows rather than twelve, a permissive YES code (under a strictly differential reading GNWT falls to 9 stated predictions, IIT to 8), one coder — are set out in Supplementary Note S6. The second independent route, coding against the union of what proponents themselves name, has not been taken, and we name it as the next check rather than claim its result.

7.6 Table 2: content inventory

The inventory asks a different question of the same theories: what content have their tests actually manipulated? Its primary unit is the publication, because the partition into experiments could be read only where a full text was available — at submission that was 18 of 42 rows carried as one publication each; after the full-text reading of 25 September 2026, 1 of 59 rows remains so carried as one row (Table 2 caption); experiment-level counts are given beside the publication counts and reproduce from the same file. The sample is purposive — the primary statements of the coded theories, the studies those statements cite as support, and the adversarial-test literature, under the deposited inclusion criteria. Four further publications — the Cogitate protocol, the taxonomy paper of the masking family, the front-versus-back review and the theory paper behind the sustained-intentions paradigm [1,119] — define paradigms rather than report data, are listed as paradigm sources, and are outside every denominator. Content is coded on three raw fields, modality, affective status and (for motor rows) effector, from which the class is derived by script; interoceptive content means content from visceral afferents or an autonomic signal modulating access, and motor conflict is split by effector, skeletal or autonomic. The category that does the analytic work is “theory-addressed”, and its rule (criterion I2) keeps three attributions apart: a study is theory-addressed when its own authors name a coded theory as tested or as the framework their result bears on, or when a later paper that is a coded theory’s primary statement cites it; the coder’s own inference is recorded separately and enters only a sensitivity scenario. Under that rule the baseline denominator is 21 publications (30 experiment-level rows), with one publication whose access was suppressed rather than manipulated excluded [129], the metacognition studies included as contested rows, and one interoceptive attention study included but contested [130].

Table 2. Content whose access was manipulated in the publications the coded theories are tested by or built on (Table S2). Counts are k of n under the baseline attribution scenario S0 of the deposited sensitivity file, at publication level (the denominator used in the text and the abstract) and at experiment-level row (in the baseline denominator of 45: 44 experiments and 1 publication carried as one row because its full text has not been obtained; the full corpus of 59 rows is 58 and 1). A publication whose rows fall in two sub-classes of motor conflict counts once under motor conflict. Of the 32 publications, 8 are not theory-addressed under the rule, 2 are state studies without a content contrast and 1 suppressed rather than manipulated access [129]; the last enters only as a sensitivity scenario.

Content class	Publications, baseline S0 (n = 23)	Experiment-level rows, baseline S0 (n = 45)	Publications, no theory filter (n = 29)
Neutral visual	16 of 23	32 of 45	18 of 29
Competing motor policies (any effector)	4 of 23	8 of 45	4 of 29
— of which autonomic effector	1 publication, as a control study beside skeletal experiments	1 of 45	as at baseline
Neutral non-visual	2 of 23	4 of 45	2 of 29
Valenced	0 of 23	0 of 45	3 of 29
Interoceptive (visceral afferent content)	1 of 23	1 of 45	2 of 29

The finding is an imbalance, stated in the terms the attribution rule licenses. Under that rule, 16 of 23 theory-addressed publications manipulated neutral visual content, 4 of 23 competing motor policies — all from the supramodular programme, one of them containing, beside its skeletal experiments, the control study in which the effector is autonomic — 2 of 23 neutral non-visual content (the auditory no-report studies [104,106]), 0 of 23 valenced content and 1 of 23 interoceptive content; at experiment-row level the same corpus gives 32 of 45, 8 of 45 (7 skeletal, 1 autonomic), 4 of 45, 0 of 45 and 1 of 45. The single interoceptive publication is the cardiac interoceptive-accuracy study that a coded theory’s primary statement cites [116], a contested inclusion; in the strict scenarios the cell is empty. The smooth-muscle control study of the supramodular programme [40], coded interoceptive in the earlier version, is autonomic-effector motor conflict, since what it measures is experienced conflict between a sustained intention and the pupillary reflex, not a visceral afferent signal; it is the only theory-addressed experiment on autonomic-effector conflict in the corpus and the empirical anchor of column M4c. The valenced cell is empty because the three valenced publications in the corpus — fearful expressions breaking continuous flash suppression faster than neutral ones [128], arousal and valence varied orthogonally [131], threat-signalling input gaining preferential access [132] — are named by no coded theory’s own authors and cited by no coded theory’s primary statement: all three have now been read in full (the threat study cites the global workspace model in its introduction without testing it, which the rule does not count as attribution), so the zero is a result of this rule on this corpus, not an established absence of attribution in the wider literature, and we withdraw the earlier formulation that no coded theory claims the valenced or interoceptive literature.

Two scenarios in the deposited sensitivity file decide how much the imbalance can carry (publication level, experiment-row value in brackets). Adding the masked-expression study whose access was suppressed [129] makes valenced content 1 of 24 (1 of 46) — the only theory-filtered scenario in which the zero does not survive. Accepting a later attribution only when its citing sentence was read and names the theory, and removing every contested row as well, gives neutral visual 8 of 14 (15 of 27), the smallest neutral-visual share in any scenario. The remaining scenarios (Supplementary Note S6.2; `s2_sensitivity.csv`) move the counts without changing the pattern, and the earlier version’s publication-level counts reproduce from the legacy codes, so the change of attribution rule and content classes, not recoding, accounts for the difference between versions. The predominance of neutral visual content survives every scenario at both units; the zero in the valenced cell rests on the full texts of the three valenced publications, read on 25 September 2026, none of which names a coded theory as tested or as the framework its result bears on; the abstract states the two together as

one imbalance at publication level under the rule named here. The pattern is corroborated from outside our coding: the field's census of 412 experiments codes stimulus modality and category as methodological variables and reports no valence breakdown at all [78]. The publication remains the primary unit because it is the unit at which the two coders' partitions can be compared and because the partition of 1 of the 32 publications is still unverified (Section 7.7).

7.7 What survives reclassification, and limitations

Because the **INTERPRETED/NOT_LOCATED** boundary carries the weight, we tested the **derived classes** against the least favourable reclassification available: promoting every **INTERPRETED** cell to **EXPLICIT**, that is, deciding every judgement call against our own conclusion. The M1 and M2 results survive in the sense that matters for Section 9. M1 has 1 **EXPLICIT** and 1 **INTERPRETED** cell (passive frame theory's constructed null), M2 likewise 1 **EXPLICIT** and 1 **INTERPRETED** (the same null; predictive processing's positive on M2 went to **NOT_LOCATED** on full-text reading); promotion would set the stated supramodular positive beside a stated passive-frame null — a pair the register would enter as discriminating on sign, making the column contested between two formulations of one programme (Section 4) — so we claim only that as coded the columns are single-occupant, the one reading that changes this promoting a null the passive-frame authors have not stated. The M5 result does not survive: promoting its four **INTERPRETED** cells turns a column with no stated prediction into 4 of 12, and a coder who read the heartbeat-evoked response as a joint cortical-autonomic measure would fill one cell without promotion (Section 7.4 gives the per-cell judgement). The M8 class is unaffected, resting on 6 **EXPLICIT** cells and the discriminating **GNWT-IIT** pair on temporal profile; the M4a–M4c classes are unaffected in the other direction, since promotion adds occupants of the same sign and creates no discriminating pair; and M6 turns on the adjudication described in Section 7.4.

A second sensitivity runs the other way. Two of the 21 **EXPLICIT** cells carry an adjudication flag: the attention schema theory on M8, whose temporo-parietal locus claim is not in the source the earlier version cited [84] but in the empirical paper the proponents cite for it [133], now read in full, where the authors themselves hedge the overlap between the attribution and attention sites, and in an earlier statement of the theory [134]; and the Feinberg–Mallatt scheme on M4a, whose explicit interoceptive category is in a book we could not read. The neural subjective frame on M4b, flagged in the earlier draft as coded from an abstract, has been read in full and the flag withdrawn: the sources state the modulation directly, with the modulator the cortical response to the heartbeat rather than cardiac phase or bodily state. Downgrading the first leaves M8 contested with 5 **EXPLICIT** cells; downgrading the second leaves M4a with two **EXPLICIT** cells, still occupied but not contested; neither creates a contested column. The 24 flagged cells are listed with their reasons in Table S1 and are the first business of the adjudication meeting. The sensitivity analysis is prepared for deposit with the coding.

Five limitations bear on how much weight Tables 1 and 2 can carry, stated without mitigation. First, the codes are a single coder's; until the blind second coding of Section 7.2 is complete, the evidential base of this section is 120 cells, 34 register entries, 63 pair relations and 42 content classifications produced by one reader applying a written scheme to texts, and we make no reliability claim. Second, the evidence base was partly second-hand at submission and has since been read: at the first revision

draft 57 of the 120 cells rested on abstracts; after the full-text reading of 25 September 2026, 118 cells rest on full text and 2 (higher-order thought theory on M6, predictive processing on M1, both NOT_LOCATED) on abstracts whose sources have not been obtained [63,75,76]. The reading changed 5 codes (3 stronger: supramodular interaction theory on M4a and M4b, recurrent processing on M7; 2 weaker: GNWT on M5, predictive processing on M7; all listed with the previous code in Table S1), left 45 unchanged and added 6 candidate re-codes to the adjudication list rather than to the table — 3 at confidence 1, 2 where the source is auxiliary or admits two readings, and 1 (passive frame theory on M2) where the candidate sentence concerns the presence of conflict rather than its degree. The founding supramodular statement [38] has now been read in full, and P16 is verified from it as well as from the later synthesis [39]. The inventory publications have likewise been read: the partition into experiments and the analysed participant counts are read from Methods for 31 of the 32 publications, and the one publication whose PDF could not be obtained (the theta-burst study [137]) is carried as one row with its partition marked unverified. Each such cell is marked in the deposited tables, and a stated prediction resting on an abstract is a weaker warrant than one resting on a quoted passage. Third, both the theory set and the study set are purposive samples; the accounts that meet the inclusion rule and were not coded are named in Section 7.1, and a registered search would very likely change the counts in Table 2, substantially so for the smaller cells. Fourth, the column set is the first author's, and the only independence check performed is that of Section 7.5, which covers ten of the twelve rows and checks occupancy rather than disagreement; the letter-to-proponents route, which would also settle whether P24 is a prediction its proponents own, has not been taken. Fifth, coding a “predictive apparatus” requires deciding which formulation of a theory counts as canonical; for theories with several revisions we coded the most recent statement making empirical commitments — defensible but not neutral, and the reason two pairs of rows that were one are now two.

8. Online metarepresentation, policy candidacy and reconstruction

The upper part of the episode — registering one's own processing, treating a proposed action as one option among several, reconstructing it afterwards — holds three dissociations that are framed but, with the partial exception conceded in Section 8.2, not measured, each conflated with a measured neighbour.

8.1 Precision, not registration

The accuracy of judgements about one's own processing varies separately from that processing's accuracy: meta- d' expresses metacognitive sensitivity in units of first-order d' [135], and calibration, error sensitivity and performance dissociate [136]. Whether theta-burst TMS to dorsolateral prefrontal cortex impairs metacognitive sensitivity with discrimination intact is disputed [137–140]; we decide nothing here, and the higher-order locus prediction in the register (Table S4, P09) carries the dispute as an auxiliary assumption. Lesion data implicate anterior prefrontal cortex, whose integrity predicts introspective rather than perceptual accuracy [141,142], with domain-general and domain-specific components [143] and a separate evaluative stage [144]. Registration grants no control, but its *precision* predicts downstream change — information search [145], resistance to counter-evidence [146], metacognitive bias shifting with symptom dimensions while performance stands [147]. The measurable variable is metarepresentational precision, not the fact of registration,

which is not the claim most theories in Table 1 make. Two qualifications: self-registration may not be self-directed — confidence compares contributions to joint decisions [148], may be a culturally acquired skill [149] secondary to mind-reading [150] or a workspace’s admission criterion [151], and in the radical version consciousness is a learned metarepresentation [152–154]; and the report scales are not equivalent — the Perceptual Awareness Scale exists because binary seen/unseen flattens intermediate states [155] and is more sensitive than confidence [156], the discreteness dispute is unresolved [157] against workspace theories predicting a threshold [158], and scale and quantity are poorly coordinated [159].

8.2 Policy candidacy

Policy candidacy (polyvariant choice) is the mode in which a proposed action policy ceases to be the sole command and becomes one member of a compared set: a change in the proposal’s status, a comparison set, and independence from both online metarepresentation and the success of control. The obvious label, *meta-control*, is taken — in computational neuroscience it means arbitration between controllers and tuning of their parameters [160–163], parametric arbitration rather than a policy represented as an object — and we do not use it.

Existing paradigms measure everything adjacent to policy candidacy and not it. Meta-d’ measures the evaluation of a decision already taken [135]. The expected value of control measures the *intensity* of allocation in dorsal anterior cingulate cortex [164,165] over a cascaded prefrontal organisation [166] whose limit is task separation, not fuel depletion [167] — the resource model failed a preregistered replication [168] and a multi-site test [169], and effort is recast as opportunity cost [170]. The stop-signal paradigm measures the speed of cancelling one prepared action [171,172], which masked signals achieve without awareness [173,174] through mechanisms dissociable from conscious control [175]; conflict detection recruits control with no commitment to consciousness [176]; intentional inhibition, where refusal is itself a choice [177,178], is the closest construct, but no alternative is generated and compared there either.

A correction strengthens the case. The first quantity operationalising policy candidacy — the size of the set under consideration — *is* measured by option generation [179–181], so the claim that none is measurable is too strong, and the metric for **option-set widening** should be borrowed from there. The second — whether a decision depends on properties of the *rejected* alternatives — has an existence proof but no measurement of the kind needed: the decoy effect shows that an asymmetrically dominated alternative nobody chooses changes the choice between the two that remain [182], but not whether the rejected option was accessed or whether the shift depends on that access; we cite it for the existence of the dependence only. The selection threshold shifting with urgency [183] is the nearest precedent for a threshold measure. Unmeasured, then, are the access-dependence of the decoy effect and whether metacognitive precision predicts option-set widening rather than correctness.

8.3 Report as write-back into the self-model

Participants report not the causes of their behaviour but plausible theories of those causes [184], including about current phenomenal content [185]; in choice blindness, switched faces went unnoticed and participants justified a choice they had not made [186,187]. Stronger still: when two of

their own moral-survey ratings were reversed by a paper-slip trick, most participants failed to notice and about half argued for the opposite position moments later [188] — an immediate result, not a measure of the altered attitude’s durability, and the write-back claim rests on the immediate result. Reconstruction is therefore not a distortion of the record but a write-back into the model of oneself: report can become a cause of the subsequent state. The same holds for the sense of agency, an inference rather than access to the motor command [189,190], and for the construction of having chosen freely [191]. Two cautions: intentional binding [192] is contested, reproduced under illusory voluntary action [193] and sensitive to phenomenological control [194]; and the Libet result must be stated exactly — the averaged readiness potential [195] is not refuted as a measurement, its causal interpretation is, since stochastic accumulation makes it an artefact of averaging [196,197]. Hence “post-hoc reconstruction” rather than “reflection”, which imports a transparency the data do not support — itself analysed as the unavailability of the model’s own modelhood to the system using it [198–200].

8.4 What is measured and what is not

The three dissociations do not stand in the same place, and the ledger is uneven rather than empty. Metarepresentational precision has validated instruments — meta- d' and its calibration variants — and no experiment that asks whether precision, rather than registration, gates access; that test can be run with what exists. Policy candidacy is measured in part: option generation already counts the set under consideration (Section 8.2), and the decoy effect shows that a decision can depend on properties of a rejected alternative, whereas whether that dependence runs through access to the rejected option has no measurement at all. Write-back has been demonstrated [188] but has not been turned into a measure that any theory is committed to. What the three share is therefore not absence of measurement but absence of a claimant: none is the dependent variable any theory in Table 1 predicts, and Section 7 documents what that costs. Each would be cheaper to test than the localisation dispute that has absorbed the field’s resources; whether any of them is tested depends on whether some theory is prepared to put a prediction on it.

9. Conclusion: the minimal measurement

The field’s problem is not a shortage of theories or of rigour. It is that the coded theories concentrate their stated predictions on one of ten measurable quantities — the cortical signature of access, where one pair of stated predictions, on temporal profile, is discriminating and where the first adversarial test accordingly separated its two targets — and have stated almost nothing about the other nine: 12 of the 27 stated predictions in the register fall in that column, and the rest of the space is not contested but, for the most part, unspoken for. Section 7.4 says what each other column can carry; two of its results drive the design: the number and incompatibility of competing action policies are single-occupant, stated by supramodular interaction theory alone and confined by construction to skeletal muscle [38,39], and in the checked sample of theory-addressed experiments, under the stated attribution rule, valenced content is absent.

One experiment converts part of that description into a result, and we state exactly which part. Hold salience, task relevance and the number of distinguishable responses fixed. Vary how many of the responses the content demands share an effector and so cannot be executed together. Measure access

with a report-independent signature, and record autonomic channels alongside it. In the report session, have the participant rate the experienced conflict itself, in the theory's own variable, after each probed response. In a separate branch, kept as an estimate, vary how many responses are demanded at all, with none sharing an effector.

The design replaces the earlier version's because of a confound that design carried: incompatibility was set by the target's feature conjunction — three rules demanding three identical keys, two identical and one different, or three different keys — so the number of distinct responses rose with incompatibility and any effect could have been attributed to response-set size or congruency instead. The supramodular sources locate the gradation of conscious conflict along the compatibility axis, not the number axis — compatible plans may co-occur without any perturbation of consciousness [39] — so the earlier design could not have tested what the theory states. Here the target always demands three distinct responses, one per rule, and only the effector mapping in force for the block decides whether they can be executed jointly; the number of policies is varied only in the separate branch at zero incompatibility, where a load account — GNWT joined to a capacity assumption it has not made, registered as the derived P25 — would predict a decrement and the authors' derived P32 an increase.

9.1 One worked condition: incompatibility at fixed count under no-report

What is tested, and by whom it was predicted. We specify one: incompatibility of licensed policies at fixed count under no-report access; one confirmatory contrast tests a one-step derivation (same condition, different outcome) from a stated prediction also tested on its own variable. The stated prediction is supramodular interaction theory's P16 — conscious conflict at two or more incompatible skeletomotor inclinations, nothing stated beyond two [38,39]. The derivation is P23, ours, one step from P16: the same condition (incompatible intentions), a different outcome (access to the masked target that licenses them, instead of experienced conflict [40]); a null bounds our derivation, not the theory. P16 itself is tested on its own variable — the experienced-conflict rating of Morsella et al. [40] — by a separately preregistered report contrast (P35). The count branch is estimation: P32 (an access threshold at two or more co-executable policies) and P25 (a load decrement), our derivations, have opposite signs; P33 (no further rise from two to three) is a bounded estimate. It is kept because the nuisance account “more levers, more access” predicts opposite signs on the two contrasts, so its estimate says whether lever number explains the confirmatory result.

Design. Three overlearned rules map a masked target's hue, position and aperture shape to three distinct responses on three three-position levers; responses combine only across levers. Incompatibility — 0, 1 or 3 pairs of demanded responses sharing a lever — is set by the mapping in force for the block. Seven presentation cells, five analysis cells: count 1 and count 2 at 0 shared pairs (one cell each, lever rotated between participants); count 3 at 0, at 1 (three pooled cells, one per rule pair sharing a lever) and at 3 shared pairs. Cells are blocked, mapping fixed per block: 16 blocks per session, two per cell and three in the two single-cell incompatibility cells. The third block does not equalise precision — about 99 usable no-report trials against about 199 in the pooled cell — but removes the floor risk and roughly halves the gap. Each session opens with a titration phase; the contrast is then frozen for all its blocks, nothing adapts within blocks, and between-session drift is reported. A probe on a quarter of trials names one rule, executed alone; joint execution is never

required. In session 1 only, after each probe response — never before, which would disrupt execution and latency — the participant rates the urge to err on the 8-point scale of Morsella et al. [40].

Trial budget. Presented and usable trials per cell, session durations (session 2 about 105 min, one visit) and the fixed $N = 120$ (119 required at $d_z = 0.30$ for one uncorrected contrast, rounded to a multiple of the six lever rotations; no sequential monitoring) are derived in Supplementary Protocol S5.

Access index and autonomic recording. Trial-wise decoded evidence for “seen” from a per-participant classifier trained on the rating-labelled session-1 no-probe trials: regularised linear discriminant analysis on 100–500 ms post-target amplitudes at 17 parieto-occipital channels, Perceptual Awareness Scale rating [155] binarised as 1 against 2–4, nested block-wise cross-validation, specification frozen before session 2 is scored [103,201–203]. The index is EEG-only, a candidate marker of access, not a measure of M5; because a no-report participant may covertly categorise [102,108], the confirmatory contrast is also computed on the session-1 rating. ECG, skin conductance and pupil are recorded throughout; a cortical–autonomic coupling index is reported per cell with estimate and interval, estimation only: no coded theory predicts over it.

Contrasts and power. One confirmatory contrast on the access index, C1a, tests P23: 0 shared pairs against the mean of 1 and 3 at count 3 (weights $-1, \frac{1}{2}, \frac{1}{2}$), $\alpha = 0.05$ two-sided, no multiplicity correction; it is supported only if its EEG-index and PAS-report versions are both significant in the predicted direction. Power at $N = 120$ is 0.90 at $d_z = 0.30$, 0.82 for the two-version rule under independence. CR tests the stated P16 on the session-1 rating, same cells and weights, in its own preregistered family, at an assumed $d_z = 0.50$ — under a third of the only anchor, $d_z = 1.66$ on the same rating [40] — with power above 0.99 (0.90 at $d_z = 0.30$). C0 (count 1 against the mean of 2 and 3 at 0 shared pairs) is an estimate with two-sided interval, its sign read against P32 and P25; the count-3 minus count-2 difference is reported with the narrowest equivalence bound it satisfies [204], P33 being supported only inside ± 0.30 d_z . C1b (linear trend, two-df omnibus) is exploratory.

Mandatory pilot. Four criteria, preregistered in S5, must be met before the confirmatory sample runs: visibility inside its band after titration and under the frozen contrast; classifier $AUC \geq 0.70$ across incompatibility levels — a validity test that defines the population addressed without selecting for the hypothesised effect; probed-rule accuracy at criterion in every cell; and a pilot urge-rating difference between the 3-shared and 0-shared cells of at least $d_z = 0.50$, the value assumed for CR, below which that test is unwarranted and the manipulation, not the hypothesis, has failed. Failure on any criterion stops the study; the pilot is reported regardless.

Valence. Study 2 repeats the design with negative, neutral and positive content, testing negative against positive at matched arousal — P34, our derivation, not the stated P13, which concerns experienced valence and free-energy change. Pupil and skin conductance, possible consequences of the manipulation, are arousal-matching checks and sensitivity-analysis covariates only; the incompatibility $\times$ valence interaction is estimated, not tested (S5). A valence effect would corroborate our derivation and separate no coded theory.

9.2 The condition that cannot yet be run, and what would make it adversarial

The earlier version presented an autonomic-only condition — a conflict resolved wholly by autonomic effectors — as the most discriminating test in principle, setting supramodular interaction theory's stated null against an interoceptive-inference positive; Section 7.4 has withdrawn the second half of that opposition, the positive being our derivation (P24). As things stand the condition tests two stated nulls and one derivation; it becomes adversarial, and M4c the first contested column outside M8, only if proponents of interoceptive inference endorse P24 or state a prediction of their own — the cell the letter-to-proponents route of Section 7.7 would most directly affect.

The design difficulty is unchanged and we flag it rather than leave it for a reviewer. Most homeostatic crises recruit skeletal effectors — respiration through the diaphragm and intercostals, thermoregulation through shivering and posture, so breath-holding is not evidence against a skeletomotor boundary. What is required is a conflict whose resolution is wholly autonomic: antagonism between thermoregulatory and gastrointestinal demands with posture fixed and respiratory pattern change blocked, or orthostatic and thermal loading of the baroreflex without postural correction. De-efferentation (high spinal injury, locked-in syndrome) is not a discriminator either, since the claim concerns intentions rather than execution and the central skeletomotor plan is preserved. Blocking respiratory pattern change is itself a skeletomotor intervention and does not escape the objection: in an awake participant, skeletomotor intentions survive any fixation of posture or breathing, and supramodular interaction theory is a claim about intentions.

The deeper obstacle is the one component of the missing measurement that is not routine (Section 7.4): a test of M4a or M4c requires a report-independent measure of access to interoceptive content — a signature tracking, trial by trial, whether an interoceptive event was accessed, validated against report in a calibration set, generalising to no-report blocks, and dissociable from afferent strength, arousal and attention to the body, which alone modulates the heartbeat-evoked potential [205,206]. The visual and auditory measures of Section 7.4 do not transfer, cardiac interoceptive accuracy measures require report by construction [116], and none has been validated for interoceptive content. The order of work is therefore the reverse of the order of interest. The condition of Section 9.1 uses established visual no-report measures and can be run now; the informative outcome for the derivation it tests (P23) is a bounded null, and it carries a direct report-based test of the stated P16 alongside. The autonomic condition would need both a validation study and a stated prediction from the other side before it could adjudicate anything. That is the actual conclusion of this review — a smaller claim than an adversarial test on a new quantity, but one that names both the measurement and the statement whose absence keeps the cell closed. Presenting the autonomic condition as ready would be the kind of overclaim this review has been criticising.

A closing note on what this review has and has not done. We have not proposed a theory. We have documented, by coding, the concentration with which this section opened, and specified the experiment built on the one stated prediction the audit found outside that quantity, which tests a one-step derivation of it under no-report access and the prediction itself on its own report variable. A theory that does make predictions about the single-occupant and thin coordinates — about the format in which incommensurable local evaluations become comparable, its dimensionality as a function of the number and heterogeneity of conflicting domains, and organism-wide reconfiguration breadth as a second condition on access, exists and is set out separately, as Arbitration Format

Theory, in a companion paper. It does not propose an alternative physical basis for consciousness, and its claims are stated over observables rather than over experience. Whether it is right is a separate question from whether the columns are single-occupant or thin. They are so either way, and the experiment that operationalises the one stated prediction among them is worth running whoever turns out to have predicted the answer.

Supplementary Materials

Supplementary Materials: The following will be deposited at [repository] under DOI [10.xxxx/PLACEHOLDER — archive assembled; DOI to be inserted on public deposit] and accompany the submission. Table S1: theory-by-quantity coding matrix (120 cells: 12 coded accounts by 10 quantities, with code, polarity, relation, per-cell justification, source DOI and locus, evidence status, confidence, provisional and adjudication flags, and the legacy code of the earlier version where one exists). Table S2: content inventory (32 publications in 59 experiment-level rows with a unit-type field, with paradigm, modality, affective status, effector, derived content class, the three attribution columns and the derived theory-addressed flag, contested-inclusion flags with reasons, and evidence status), with `paradigm_sources.csv` (the four paradigm-defining publications and the rows that rely on them) and `s2_sensitivity.csv` (every scenario reported in Section 7.6). Table S3: coding of ten theory rows against the twelve ConTraSt annotation dimensions (120 cells). Table S4: register of 36 entries (27 stated predictions, 2 kept for provenance after re-coding to INTERPRETED, 6 derived by the authors and labelled so) with source, locus, condition, outcome measure, expected result and confidence; `pair_register.csv`: every cross-theory pair of stated predictions in every column with two or more (63 pairs) with shared condition, shared observable, four-way relation, favouring results and pair-level confidence. Supplementary Protocol S5: full protocol of Study 1 and Study 2 (stimuli, task, effector mapping, cell list, presented and usable trials, exclusions, index freezing, autonomic channels and coupling metric, contrasts and weights, power tables, pilot criteria). Supplementary Note S6: the ConTraSt column set of Section 7.5 (provenance, occupancy by dimension, correspondence to M1–M8, limits) and every Table 2 sensitivity scenario of Section 7.6. Codebook v2 (frozen; version and hash printed in its footer), the inclusion criteria for Tables S1 and S2, the blank second-coder forms, and the scripts that produce every count in Section 7 and every power figure in Section 9 (`column_typology_pairs.py`, `build_table2.py`, `s2_sensitivity.py`, `agreement.py`, `section9_power.py`, `protocol_numbers.py`, `reproduce.py`).

Data Availability Statement

Data Availability Statement: Tables S1–S4, Supplementary Protocol S5, the codebook, the inclusion criteria and the scripts will be deposited at [repository] under DOI [10.xxxx/PLACEHOLDER]; the archive is assembled and the DOI will be inserted on public deposit. No new empirical data were generated.

Declarations

Author Contributions: Conceptualisation, [Author 1]; methodology, [Author 1] and [Author 2]; investigation (first coding of Tables S1–S4 and of the content inventory), [Author 1]; validation (blind

second coding of Tables S1 and S2 against the frozen codebook — scheduled; its agreement statistics will be inserted in Section 7.2 when complete), [Author 2]; software (typology, agreement and power scripts), [Author 1]; writing – original draft, [Author 1]; writing – review and editing, [Author 1] and [Author 2]. Both authors have read and agreed to the published version of the manuscript.

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Institutional Review Board Statement: Not applicable; no new empirical data were collected. The studies specified in Section 9 and Supplementary Protocol S5 are proposals and will be submitted for ethical review before any data collection.

Informed Consent Statement: Not applicable.

Use of Generative AI: The authors used two generative AI systems — Claude (Anthropic) and ChatGPT (OpenAI) — under their direction, for literature search and retrieval, for extracting candidate passages from sources and proposing draft codes for the coding tables, for drafting and editing text, and, in the case of the second system, for independent auditing of the deposited tables, scripts and text; a third, separately instructed session of the first system produced the independent AI coding of Table S1 reported in Section 7.2, which is deposited and is not the second author’s coding. Every code that remains in Tables S1–S4 was checked by the first coder against the full text or the abstract of its source, as marked in the evidence-status field; every count in Sections 7 and 9 is produced by the deposited scripts from the tables; all cited sources were verified against CrossRef and DataCite records; and the authors take full responsibility for the content of the manuscript.

Conflicts of Interest: The authors declare no conflicts of interest. [Authors: the manuscript argues at length with supramodular interaction theory and names a companion theory of the first author’s; any professional relationship with the proponents of the coded theories, and the relationship to the companion paper, belong here.]

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